ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording. Symbols and circuits may be drawn in different but correct ways.

QUESTION 1

1.1 Any FOUR ✓✓✓✓: electrical; mechanical (kinetic / potential); heat (thermal); light; sound; chemical; nuclear. (4)

1.2.1 Second-class lever (the load is between the fulcrum and the effort) ✓ (1)

1.2.2 $\text{Effort} \times 1{,}2 = 400 \times 0{,}4$ ✓; $\text{effort} = 160 \div 1{,}2$ ✓ $= 133{,}3\ \text{N}$ ✓ (3)

1.2.3 $\text{MA} = \text{load} \div \text{effort} = 400 \div 133{,}3$ ✓ $= 3$ ✓ (2)

1.3 $N_2 = N_1 \times T_1 \div T_2$ ✓ $= 60 \times 18 \div 54$ ✓ $= 20\ \text{r/min}$ ✓ (3)

1.4 A clutch connects and disconnects the drive between a motor (engine) and the load ✓, so the load can be started and stopped smoothly while the motor keeps running ✓. (2)

1.5 Torque is the turning effect of a force ✓ = force × perpendicular distance from the turning point ✓; unit newton metre (N·m) ✓. (3)

1.6 $T = F \times r = 50 \times 0{,}3$ ✓ $= 15\ \text{N·m}$ ✓ (2)

[20]

QUESTION 2

2.1.1 OR function ✓ (1)

2.1.2 E1 = (A + B) · C ✓✓ (2)

2.1.3

A	B	C	E1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

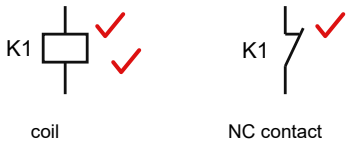
(eight rows of A, B and C ✓; E1 = 1 only when C = 1 and A or B (or both) = 1 ✓✓✓) (4)

2.2.1 $P = V \times I = 24 \times 0{,}5$ ✓ $= 12\ \text{W}$ ✓ (2)

2.2.2 $R = V \div I = 24 \div 0{,}5$ ✓ $= 48\ \Omega$ ✓ (2)

2.3 A push button only stays operated while it is pressed and springs back when released (momentary) ✓✓. A latching switch stays in its new position after it is operated until it is operated again ✓✓. (4)

2.4



Coil: rectangle labelled K1 ✓✓; NC contact labelled K1 ✓ (3)

2.5 It measures voltage, current and resistance (and continuity) ✓✓ in electrical circuits. (2)

[20]

QUESTION 3

3.1.1 5/2-way valve ✓, pilot-operated on both sides (double pilot valve) ✓ (2)

3.1.2 It has no spring ✓, so it stays (remembers) in the last position it was switched to ✓ even after the pilot signal is removed, until a signal on the other side switches it back ✓. (3)

3.1.3 Pressing 1.2 sends a pilot signal to port 14 ✓; valve 1.1 switches and the cylinder extends ✓; it stays extended when 1.2 is released (memory) ✓; pressing 1.3 sends a signal to port 12, valve 1.1 switches back and the cylinder retracts ✓. (4)

3.2 Direct control: the cylinder is controlled straight from the valve that the operator presses (suitable for small cylinders) ✓✓. Indirect control: the operator's valve sends a small pilot signal to a main (power) valve, which controls the cylinder (used for large cylinders or long distances) ✓✓. (4)

3.3 $A = \pi \times (D^2 - d^2) \div 4$ ✓ $= \pi \times (0{,}04^2 - 0{,}016^2) \div 4$ ✓ $= 1{,}056 \times 10^{-3}\ m^2$ ✓; $F = p \times A = 600\,000 \times 1{,}056 \times 10^{-3}$ ✓ $= 633{,}3\ N$ ✓ (5)

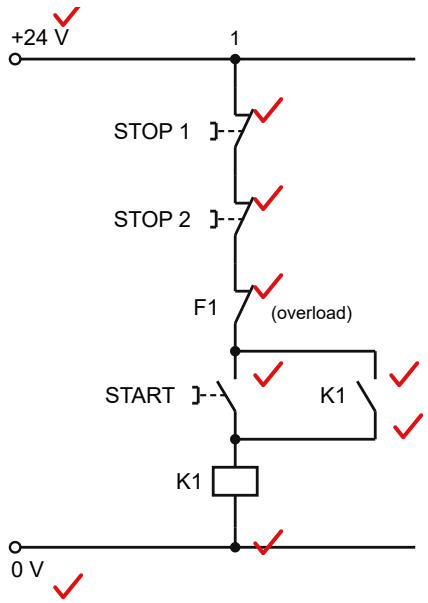
3.4 5 = the number of ports (connections) ✓; 2 = the number of switching positions ✓ (2)

[20]

QUESTION 4

4.1 Any FOUR ✓✓✓✓: stator / field magnets (or field winding); rotor / armature; commutator; brushes; shaft; bearings; frame (yoke). (4)

4.2



Supply lines ✓; STOP 1 (NC) ✓ and STOP 2 (NC) ✓ in series; overload F1 (NC) in series ✓; START (NO) ✓ with a K1 holding contact in parallel ✓✓; coil K1 ✓; correct symbols ✓ and labels ✓

(10)

4.3 So that pressing it (or a broken wire) opens the circuit and stops the machine ✓ – it is fail-safe ✓. (2)

4.4 AC changes its direction (and size) continuously, e.g. 50 times a second ✓✓. DC flows in one direction only at a steady value, e.g. from a battery ✓✓. (4)

[20]

QUESTION 5

5.1.1 Red – danger / stop / fire equipment ✓

5.1.2 Yellow – caution / warning ✓

5.1.3 Blue – mandatory instruction (must do, e.g. wear PPE) ✓

5.1.4 Green – safe condition / first aid / emergency exit ✓

(4)

5.2 Any FOUR ✓✓✓✓: electric shock; moving and rotating parts; compressed air; hot surfaces (soldering irons); sharp edges; noise; slippery floors; flying particles. (4)

5.3 Any FOUR ✓✓✓✓: plasters; bandages; sterile gauze / dressings; disposable gloves; antiseptic; scissors; burn gel; eye wash; CPR mouthpiece. (4)

5.4 Fault-finding is a logical, step-by-step way of finding the cause of a fault ✓✓. Any TWO methods ✓✓: half-split method; input-to-output method; output-to-input method; substitution method. (4)

5.5 Any FOUR ✓✓✓✓: dirt causes wear of moving parts; dust blocks cooling and causes overheating; clean machines make leaks and damage easy to see; prevents sensors from giving false signals; improves safety (no slippery oil); longer machine life. (4)

[20]

TOTAL: 100

END OF MARKING GUIDELINE